

Naive Bayes

Lecture 18

Bayes Theorem

$$P(A \mid B) = \frac{P(A) \times P(B \mid A)}{P(B)}, \quad P(B) \neq 0$$

$$P(h \mid D) = \frac{P(h) \times P(D \mid h)}{P(D)}, \quad P(D) \neq 0$$

Naive Bayes – Example 1

Ser	Color (X)	Fruit (y)
1	red	apple
2	red	apple
3	green	apple
4	green	orange
5	orange	orange
6	red	apple
7	green	apple
	...	...
	...	...

Step 1: Create contingency or frequency table for each feature of training data (Using Panda's crosstab() method). Since we have only one feature, which is color, so. we only need to build on frequency table as shown below:

		y		
		apple	orange	Total
X	red	33	0	33
	green	20	7	27
	orange	1	39	40
	Total	54	46	100

Naive Bayes – Example 1

Step 2: Compute the **marginal** probabilities for each of the output label (apple, orange):

$$P(y = \text{apple}) = \frac{54}{100}$$

$$P(y = \text{orange}) = \frac{46}{100}$$

Step 3: Compute the **marginal** probabilities for the only input feature:

$$P(X = \text{red}) = \frac{33}{100}$$

$$P(X = \text{green}) = \frac{27}{100}$$

$$P(X = \text{orange}) = \frac{40}{100}$$

Step 4: Compute the **conditional** probabilities:

$$P(X = \text{green} \mid y = \text{apple}) = \frac{20}{54}$$

$$P(X = \text{green} \mid y = \text{orange}) = \frac{7}{46}$$

Step 5: Substitute all the probabilities into the **Naïve Bayes' Formula** :

$$P(y = \text{apple} \mid X = \text{green}) = \frac{P(y=\text{apple}) \times P(X=\text{green} \mid y=\text{apple})}{P(X=\text{green})} =$$

Naive Bayes – Zero Frequency Problem

- Suppose we are given a new instance of fruit having red color this time.
- In the frequency table, if you observe that out of 46 oranges you have zero orange having red color.
- Following the above described steps, we can use the Naïve Bayes' Classifier formula:

		y		Total
		apple	orange	
X	red	33	0	33
	green	20	7	27
	orange	1	39	40
	Total	54	46	100

$$P(y = \text{apple} \mid X = \text{red}) = P(y = \text{apple}) \times P(X = \text{red} \mid y = \text{apple})$$

Naive Bayes – Laplace Smoothing

- Laplace Smoothing is usually applied by adding one count to the numerator.
- And adding number of possible values of features to the denominator.
- In this example number of possible values of the only input feature is 3 (red, green, orange). So the probability after Laplace smoothing is:

		y		Total
		apple	orange	
X	red	33	0	33
	green	20	7	27
	orange	1	39	40
	Total	54	46	100

$$P(y = \text{apple} \mid X = \text{red}) = P(y = \text{apple}) \times P(X = \text{red} \mid y = \text{apple})$$

Naive Bayes – Example 2 (Multiple Input Features)

	X			y
No	Outlook (x1)	Temperature (x2)	Humidity (x3)	Rain
1	Cloudy	Cool	High	yes
2	Sunny	Mild	High	no
3	Cloudy	Hot	Normal	no
4	Sunny	Cool	Normal	no
5	Sunny	Hot	Low	no
6	Cloudy	Mild	Normal	yes
7	Cloudy	Hot	High	yes
8	Cloudy	Cool	Low	no
9	Sunny	Cool	High	yes
10	Sunny	Hot	High	no
11	Cloudy	Hot	Low	no
12	Cloudy	Cool	Normal	yes
13	Sunny	Mild	Low	no
14	Cloudy	Cool	Low	yes
15	Sunny	Mild	Low	no

Naive Bayes – Example 2 (Multiple Input Features)

$$P(y = k \mid X_i) \propto P(y = k) \times P(X_i \mid y = k)$$

$$P(y = k \mid x_1, x_2, x_3, \dots, x_m) \propto P(y = k) * \prod_{i=1}^m P(x_i \mid y = k)$$

Naive Bayes – Example 2 (Multiple Input Features)

- Let us assume that on a particular day the outlook is **cloudy** , temperature is **cool** and humidity is **normal** .
- Given the training dataset, build a Naïve Bayes' Classifier, so that given a new set of input features we should be able to predict if it will rain on that day?
- Mathematically, we need to compute and later compare the following two conditional probabilities, and whichever, probability is greater, we predict accordingly.

X				y
No	Outlook (x1)	Temperature (x2)	Humidity (x3)	Rain
1	Cloudy	Cool	High	yes
2	Sunny	Mild	High	no
3	Cloudy	Hot	Normal	no
4	Sunny	Cool	Normal	no
5	Sunny	Hot	Low	no
6	Cloudy	Mild	Normal	yes
7	Cloudy	Hot	High	yes
8	Cloudy	Cool	Low	no
9	Sunny	Cool	High	yes
10	Sunny	Hot	High	no
11	Cloudy	Hot	Low	no
12	Cloudy	Cool	Normal	yes
13	Sunny	Mild	Low	no
14	Cloudy	Cool	Low	yes
15	Sunny	Mild	Low	no

$$P(y = \text{yes} \mid x_1 = \text{cloudy}, x_2 = \text{cool}, x_3 = \text{normal})$$

$$P(y = \text{no} \mid x_1 = \text{cloudy}, x_2 = \text{cool}, x_3 = \text{normal})$$

Naive Bayes – Example 2 (Multiple Input Features)

Contingency Table for Outlook

x_1	y		
	yes	no	Total
Cloudy	5	3	8
Sunny	1	6	7
Total	6	9	15

Contingency Table for Temperature

x_2	y		
	yes	no	Total
Cool	4	2	6
Mild	1	3	4
Hot	1	4	5
Total	6	9	15

Contingency Table for Humidity

x_3	y		
	yes	no	Total
Low	1	5	6
Normal	2	2	4
High	3	2	5
Total	6	9	15

Step 1: Create contingency or frequency table for each feature of training data (Using Panda's `crosstab()` method). Since we have three features (outlook, temperature and humidity), so we need to construct three frequency tables as shown below:

Step 2: Compute the probabilities for each of the output label (yes, no):

$$P(y = \text{yes}) = \frac{6}{15}$$

$$P(y = \text{no}) = \frac{9}{15}$$

Naive Bayes – Example 2 (Multiple Input Features)

Step 3: Compute the conditional probabilities:

$$P(x_1 = \textit{cloudy} \mid y = \textit{yes}) = \frac{5}{6}$$

$$P(x_2 = \textit{cool} \mid y = \textit{yes}) = \frac{4}{6}$$

$$P(x_3 = \textit{normal} \mid y = \textit{yes}) = \frac{2}{6}$$

$$P(x_1 = \textit{cloudy} \mid y = \textit{no}) = \frac{3}{9}$$

$$P(x_3 = \textit{normal} \mid y = \textit{no}) = \frac{2}{9}$$

$$P(x_2 = \textit{cool} \mid y = \textit{no}) = \frac{2}{9}$$

Naive Bayes – Example 2 (Multiple Input Features)

Step 4: Substitute all the probabilities into the Naïve Bayes' Formula:

$$P(y = \text{yes} \mid x_1 = \text{cloudy}, x_2 = \text{cool}, x_3 = \text{normal})$$

$$= P(y = \text{yes}) * P(x_1 = \text{cloudy} \mid y = \text{yes}) * P(x_2 = \text{cool} \mid y = \text{yes}) * P(x_3 = \text{normal} \mid y = \text{yes})$$

$$= \frac{6}{15} * \frac{5}{6} * \frac{4}{6} * \frac{2}{6} = \frac{240}{3240} = 0.074$$

$$P(y = \text{no} \mid x_1 = \text{cloudy}, x_2 = \text{cool}, x_3 = \text{normal})$$

$$= P(y = \text{no}) * P(x_1 = \text{cloudy} \mid y = \text{no}) * P(x_2 = \text{cool} \mid y = \text{no}) * P(x_3 = \text{normal} \mid y = \text{no})$$

$$= \frac{9}{15} * \frac{3}{9} * \frac{2}{9} * \frac{2}{9} = \frac{108}{10935} = 0.009$$

Since,

$$P(y = \text{yes} \mid x_1 = \text{cloudy}, x_2 = \text{cool}, x_3 = \text{normal}) > P(y = \text{no} \mid x_1 = \text{cloudy}, x_2 = \text{cool}, x_3 = \text{normal})$$

therefore, we predict that it will rain

Naive Bayes – Example 3 (Text Data)

Document	Label
I am happy because I am playing cricket.	Positive
I am happy, not sad.	Positive
I am sad, I am not playing cricket.	Negative
I am sad, not happy.	Negative

- Given a new sentence I am playing cricket and happy , Can you classify it as Positive or Negative .

	y		
	Positive (+)	Negative (-)	Total
I	3	3	6
am	3	3	6
happy	2	1	3
because	1	0	1
playing	1	1	2
cricket	1	1	2
sad	1	2	3
not	1	2	3
Total	13	13	26

Naive Bayes – Example 3 (Text Data)

Step 2: Compute the probabilities for each of the output label (Positive, Negative):

$$P(y = +) = \frac{1}{2} \qquad P(y = -) = \frac{1}{2}$$

Step 3: Compute the conditional probabilities:

$$\begin{aligned} P(x_1 = I \mid y = +) &= \frac{3}{13} & P(x_2 = am \mid y = +) &= \frac{3}{13} \\ P(x_3 = playing \mid y = +) &= \frac{1}{13} \\ P(x_4 = cricket \mid y = +) &= \frac{1}{13} & P(x_5 = happy \mid y = +) &= \frac{2}{13} \end{aligned}$$

$$\begin{aligned} P(x_1 = I \mid y = -) &= \frac{3}{13} & P(x_2 = am \mid y = -) &= \frac{3}{13} \\ P(x_3 = playing \mid y = -) &= \frac{1}{13} \\ P(x_4 = cricket \mid y = -) &= \frac{1}{13} & P(x_5 = happy \mid y = -) &= \frac{1}{13} \end{aligned}$$

Naive Bayes – Example 3 (Text Data)

Step 4: Substitute all the probabilities into the Naïve Bayes' Formula:

- $P(y = + \mid X) = \frac{1}{2} * \frac{3}{13} * \frac{3}{13} * \frac{1}{13} * \frac{1}{13} * \frac{2}{13} = 0.000024$

- $P(y = - \mid X) = \frac{1}{2} * \frac{3}{13} * \frac{3}{13} * \frac{1}{13} * \frac{1}{13} * \frac{1}{13} = 0.0000121$

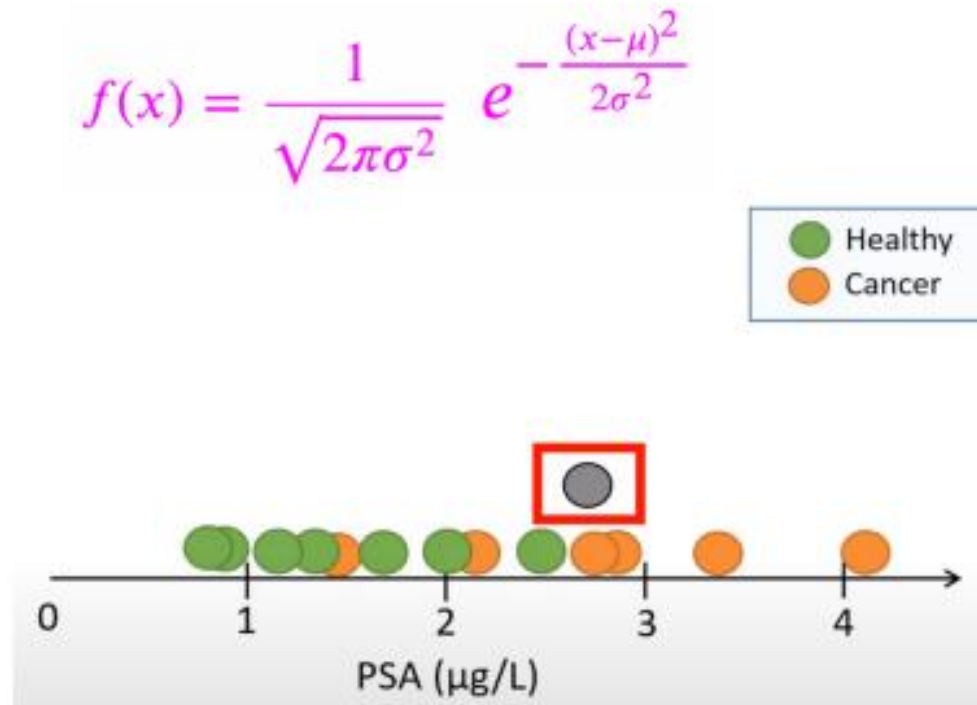
Naive Bayes – Example 4 (Continuous Input Features)

- Consider this hypothetical data set showing the measures of the Prostate-Specific Antigen (PSA) concentration from blood samples of fourteen persons, out of which seven are patients with prostate cancer while seven are healthy.
- Dataset has one input feature that is PSA level which is a continuous variable, while the output label has two possible values Cancer and Healthy.
- And the objective is to build a Naïve Bayes Classifier to predict if a given patient suffers from cancer or not given his/her PSA value.
- 100\$ Question : How to compute the probabilities when X is a continuous variable?

No	PSA	Status
1	4.1	Cancer
2	3.4	Cancer
3	2.9	Cancer
4	2.8	Cancer
5	2.7	Cancer
6	2.1	Cancer
7	1.6	Cancer
8	2.5	Healthy
9	2.0	Healthy
10	1.7	Healthy
11	1.4	Healthy
12	1.2	Healthy
13	0.9	Healthy
14	0.8	Healthy

Naive Bayes – Example 4 (Continuous Input Features)

- Answer : Use Gaussian Naïve Bayes model, which assumes that the continuous input feature variables follow a Gaussian (Normal) distribution.



No	PSA	Status
1	4.1	Cancer
2	3.4	Cancer
3	2.9	Cancer
4	2.8	Cancer
5	2.7	Cancer
6	2.1	Cancer
7	1.6	Cancer
8	2.5	Healthy
9	2.0	Healthy
10	1.7	Healthy
11	1.4	Healthy
12	1.2	Healthy
13	0.9	Healthy
14	0.8	Healthy

Naive Bayes – Example 4 (Continuous Input Features)

Calculate Sample Mean and Standard Deviation of Healthy group:

- Healthy Group:

$$\mu_{healthy} = \frac{\sum_{i=1}^n x_i}{n} = 1.5$$

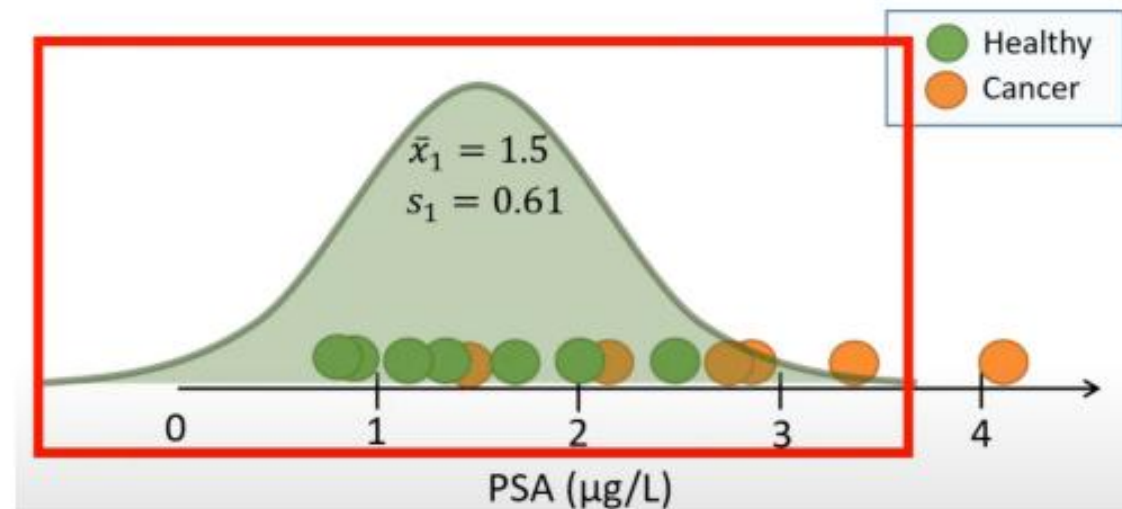
$$\sigma_{healthy} = \sqrt{\frac{\sum (x_i - \mu)^2}{n-1}} = 0.61$$

No	PSA	Status
1	4.1	Cancer
2	3.4	Cancer
3	2.9	Cancer
4	2.8	Cancer
5	2.7	Cancer
6	2.1	Cancer
7	1.6	Cancer
8	2.5	Healthy
9	2.0	Healthy
10	1.7	Healthy
11	1.4	Healthy
12	1.2	Healthy
13	0.9	Healthy
14	0.8	Healthy

Naive Bayes – Example 4 (Continuous Input Features)

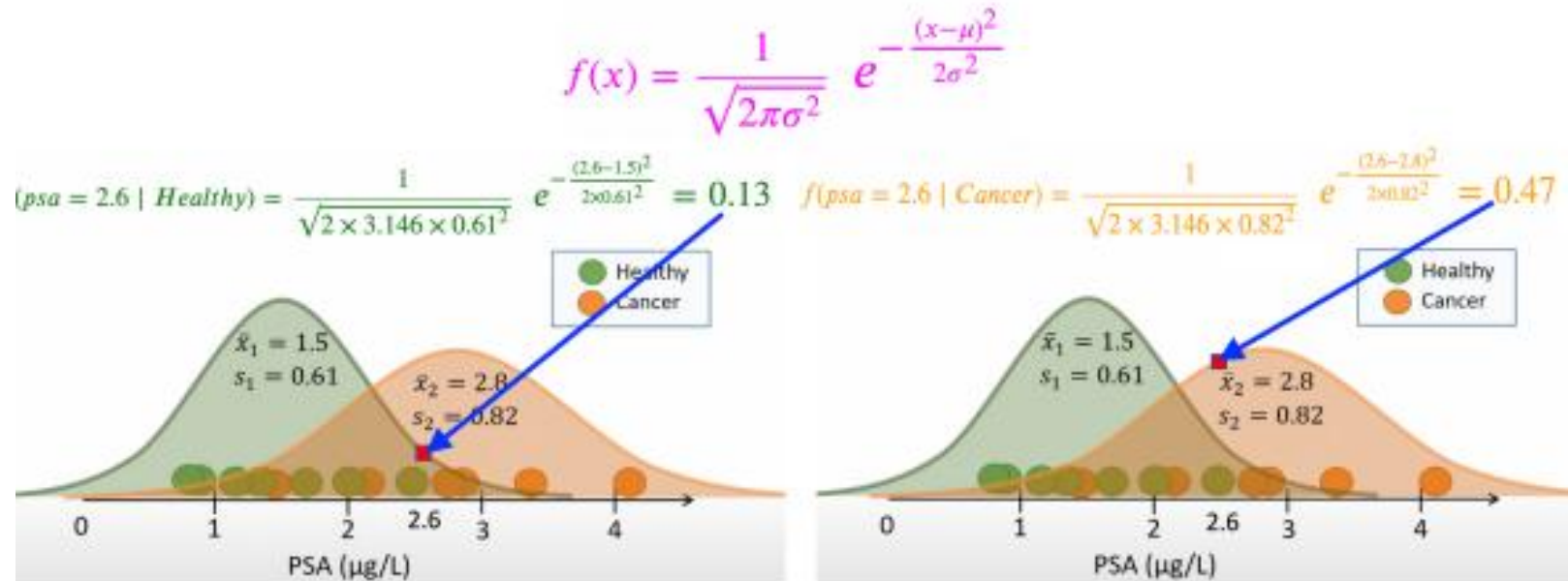
- If we plug in the $\mu_{healthy}$ and $\sigma_{healthy}$ values for the healthy group with different x_i values of PSA in the probability density function of Gaussian Distribution, we get different values which if plotted will give the following bell shaped curve for Healthy persons:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



Naive Bayes – Example 4 (Continuous Input Features)

- Healthy Group: $\mu_{healthy} = 1.5$, $\sigma_{healthy} = 0.61$
- Cancer Group: $\mu_{cancer} = 2.8$, $\sigma_{cancer} = 0.82$
- Now we need to calculate $f(psa = 2.6 | Healthy)$, and $f(psa = 2.6 | Cancer)$ for which we will use the probability density function of Gaussian Distribution as shown below:



Naive Bayes – Example 4 (Continuous Input Features)

These values 0.13 and 0.47 are not the probabilities, rather are the values of the probability density function for the height of these curves where PSA=2.6

- Substitute these values in the Naïve Bayes' Formula:

$$P(y = \textit{Healthy} \mid \textit{psa} = 2.6) = P(y = \textit{Healthy}) * P(\textit{psa} = 2.6 \mid y = \textit{Healthy})$$

$$P(y = \textit{Cancer} \mid \textit{psa} = 2.6) = P(y = \textit{Cancer}) * P(\textit{psa} = 2.6 \mid y = \textit{Cancer}) =$$

Since,

$$P(y = \textit{Cancer} \mid \textit{psa} = 2.6) > P(y = \textit{Healthy} \mid \textit{psa} = 2.6)$$

therefore, we classify the person has Cancer.

Thank You..!